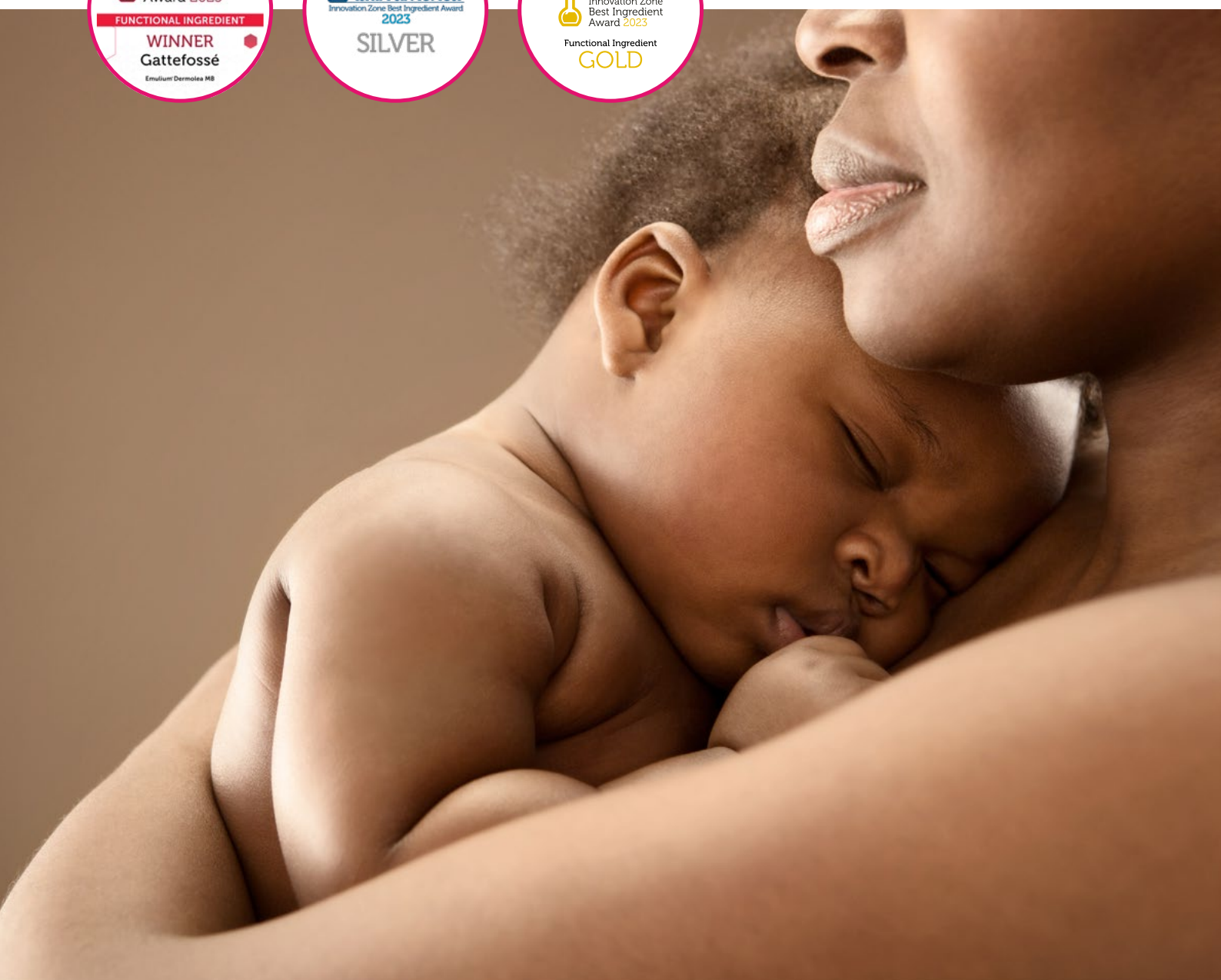


Emulium® Dermolea MB

Sensitive skin shield



People make our name



An optimal barrier function for a healthy skin



Exposed to environmental stress, the skin's surface lipids are highly impacted. The skin barrier function is altered, facilitating the penetration of irritants and increasing trans epidermal water loss (TEWL). As a result, the skin becomes dry and presents sensitivity, associated with redness and a stinging sensation. As well as general discomfort, a damaged skin barrier can lead to a burning sensation upon application of cosmetics.

There is a growing need for mild, soothing ingredients base that reduce skin sensitivity to aggression and enable the use of skincare and make-up products without reaction and discomfort.

Plant-origin O/W emulsifier

POLYGLYCERYL-6 DISTEARATE

Emulsifying properties

(and)

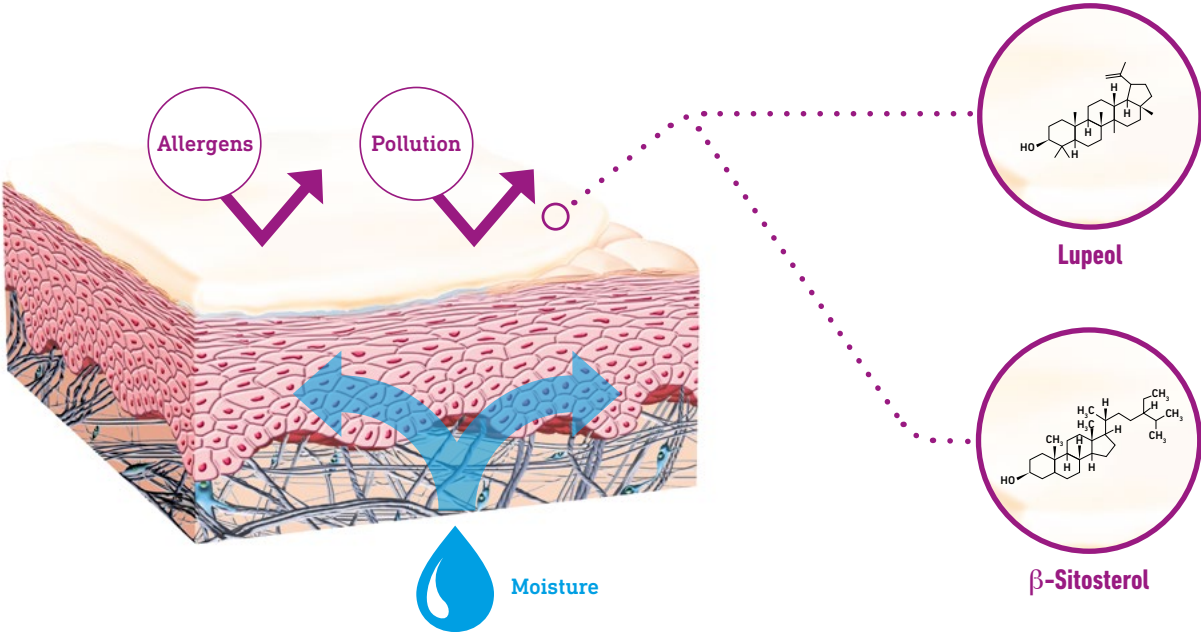
CANDELILLA/JOJOBA/RICE BRAN
POLYGLYCERYL-3 ESTERS

Sensoriality
Efficacy
Stability

A unique lipidic composition

Composed of plant lipids close to those of the skin's surface, rich in anti-inflammatory **pentacyclic triterpenes** and **sterols**, Emulium® Dermolea MB forms a fine skin-compatible film, proven to reduce environmental skin stress. Highly moisturizing and protective, it **reinforces the skin barrier function** and brings **long-term soothing**.

Wax derivatives composition		Skin's surface lipids	
Alkanes (C29-C33)	10%	Squalene (C30)	
Aliphatic esters (C36-C64)	50-60%	Wax esters (C26-46)	
Polyglycerol esters	20-25%	Free fatty acids (C12-30)	
Fatty acids/Fatty alcohols (C20-C32)	3-5%	Cholesterol	
Sterols & triterpenes	1-2%		



Multifunctional and easy-to-use emulsifier

Capable of emulsifying oil phases up to 40%, Emulium® Dermolea MB can create sprays to thick butters suitable for skin care, suncare, makeup and hair/scalp care. In the form of pellets, it is very easy to formulate and forms stable emulsions. The use of a gelling agent is recommended for better stability, particularly at high temperatures.

Direct emulsification - Hot process

70°C

Oil phase + Emulium® Dermolea MB

70°C

2000-2500 rpm

Water phase

O/W emulsion

Use level: 2-4%

1

Add Emulium® Dermolea MB to the oil phase and heat to 70°C

2

Once homogeneous, slowly add the oil phase to the heated aqueous phase under high shear (using rotor stator or a dispersing mixer)

3

Maintain high shear mixing until it is perfectly homogeneous (approx. 10-20 minutes)

4

Cool under mixing

Compatible with all kinds of cosmetic ingredients

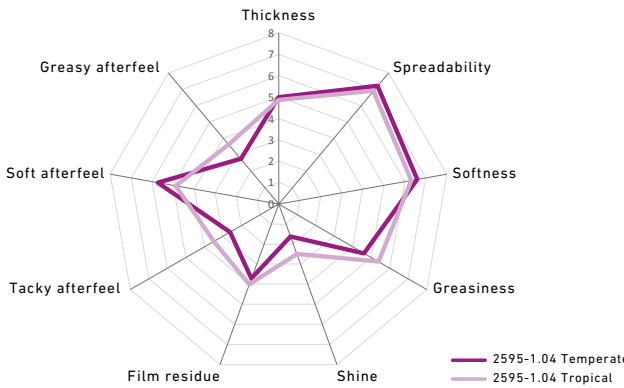
Oils	All types up to 40%
Consistency agents	All types
Gelling agents	Most gelling agents (use of anionic co-emulsifier recommended with acrylate polymers and copolymers that are not pre-neutralized)
Actives	At a pH between 4.5-9
UV filters	Compatible with organic UV filters Compatible with ZnO and TiO ₂ : addition of a high HLB co-emulsifier is recommended
Pigments	Prefer hydrophilic coatings
Electrolytes	Compatibility depends mostly on gelling agent (addition post emulsification is recommended)
Alcohol & glycols	Highly compatible
Preservatives	No known incompatibilities (except sodium anisate)

Thin and light textures

- Glossy appearance
- Easy spreadability
- Low film residue
- Low greasiness
- Smooth and comfortable afterfeel

Adapted to all climates

Similar sensory profile in temperate and tropical humid conditions



A wide range of applications

TEXTURES

From sprayable to thick textures

SENSITIVE SKIN FRIENDLY

Dermatological brands, sensitive and mature skin products

FOR ALL SEGMENTS

Skincare, suncare, haircare, scalp care, makeup

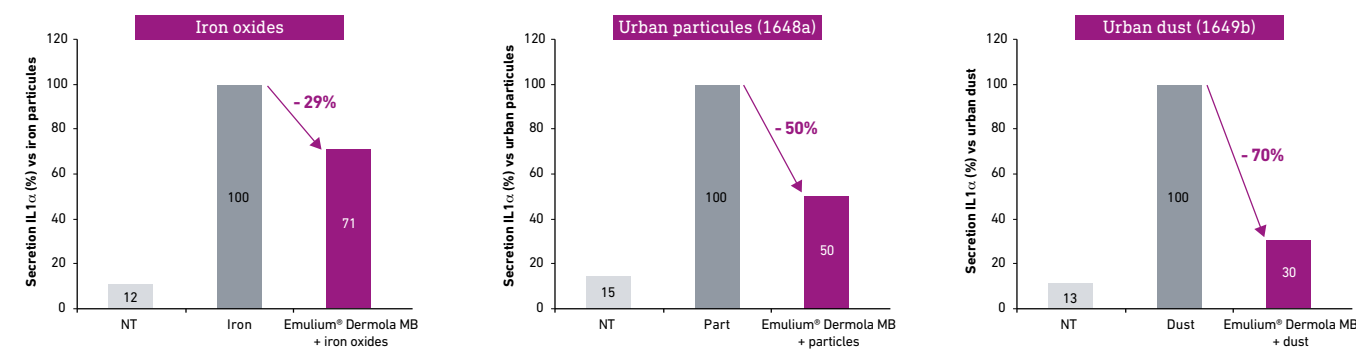
SUSTAINABILITY

Organically certified, vegan and biodegradable formulations

Durable moisturization and comfort to all skin types, even the most sensitive

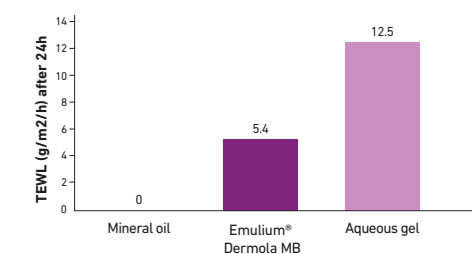
Protection against pollution

An in-house *ex vivo* test, using skin biopsies, has been performed with a cream containing 4% of Emulium® Dermolea MB to assess its effect on the secretion of IL-1 α , a marker of inflammation secreted by corneocytes when the skin is exposed to inflammatory stress.



Emulium® Dermolea MB in emulsion forms a protective shield against inflammatory stress induced by pollution particles.

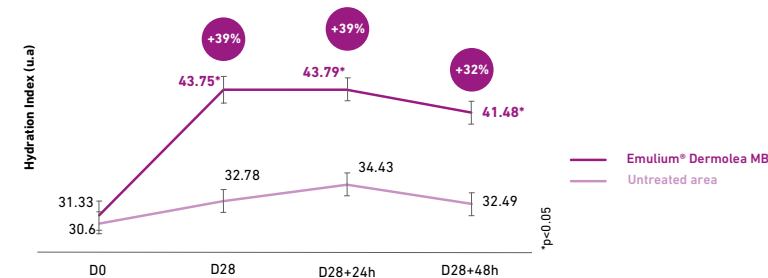
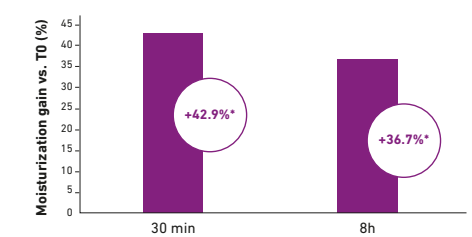
Skin barrier function reinforced



The TEWL has been measured using the Franz cell device equipped with a synthetic skin membrane over a period of 24 hours. Emulium® Dermolea MB in a cream was compared to an aqueous gel and to mineral oil.

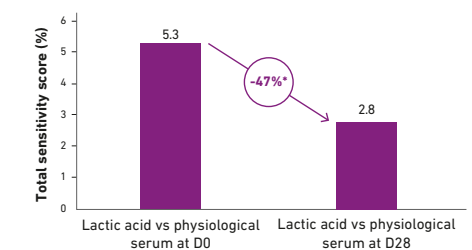
Emulium® Dermolea MB at 4% in emulsion reinforces the skin barrier function.

Long-lasting moisturization



A cream containing 4% of Emulium® Dermolea MB brings a moisturizing effect after a single application, confirmed after 28 days and this effect lasts up to 48h after the last application.

Reduced skin sensitivity



A long-term stinging test has been performed on 22 Caucasian volunteers with sensitive skin and stinger positive status (sensitivity score > 3). The skin sensitivity level has been assessed before and after twice daily application of a cream with 4% Emulium® Dermolea MB during 4 weeks.

Emulium® Dermolea MB in emulsion brings a long-term soothing effect.

Emulium® Dermolea MB

is a plant-based O/W emulsifier, easy to formulate and multifunctional, which makes it possible to formulate over a wide range of cosmetic applications.

Emulium® Dermolea MB is obtained by transesterification and functionalization of rice bran, jojoba and candelilla waxes. It forms a fine skin-compatible film, proven to reduce environmental skin stress and to increase the skin's tolerance level.

It formulates delightful textures with a light after-feel, adapted to all skin types and climates. Less sensitive to aggression, the skin is more able to receive cosmetic treatments and make-up without reaction and discomfort.

COSMOS approved and readily biodegradable, it can be used in organically certified, vegan and biodegradable formulations.

Characteristics

- O/W emulsifier
- Pellet form
- PEG-free, 100% natural origin
- HLB = 8.5

Applications

- From sprayable to thick textures
- Skin care
- Color cosmetics
- Sun care
- Hair and scalp care
- Organically certified, vegan and readily biodegradable formulations

Use level

2-4%

INCI name: Polyglyceryl-6 Distearate (and) Candelilla/Jojoba/Rice Bran Polyglyceryl-3 Esters

Worldwide approval for cosmetic applications

Listed in IECIC (Inventory of Existing Cosmetic Ingredients in China) 2021

COSMOS approved

Readily biodegradable

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